Day 7 Task Report

Title: Semantic Search System (Embeddings)

TASK-07: Building a Vector-Based Search Engine using Sentence Embeddings

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Objective

The objective of this task is to build a semantic search engine that understands the meaning of a query instead of just matching exact words.

In earlier tasks, the search system worked based on keyword matching. In this task, we improve it by using embeddings, so that even if the exact word is not present, the system can still find relevant results.

For example, searching for *"processor"* should return results related to *CPU* or *GPU*.

Tools Used

- VS Code
- Git & GitHub
- MS Word

Libraries Used

- sentence-transformers → To generate embeddings
- scikit-learn → To calculate cosine similarity

Concepts Used

1. Semantic Search

A search technique that understands the meaning of text instead of exact keyword matching.

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2. Sentence Embeddings

Converting sentences into numerical vectors so that machines can understand their meaning.

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3. Cosine Similarity

A mathematical method used to measure similarity between two vectors.

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4. Vector-Based Search

Searching data by comparing vector representations instead of text.

Code Explanation

The program works in the following steps:

1. Read sentences from corpus.txt file
2. Convert all sentences into embeddings using the model
3. Take user query as input
4. Convert query into embedding
5. Compare query with all sentences using cosine similarity
6. Display top 3 most similar results with scores

Errors Faced

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1. Module Not Found Error

Error:

ModuleNotFoundError: No module named 'sentence_transformers'

Solution:
Installed required library using:

pip install sentence-transformers

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2. Virtual Environment Issue

Problem:
Library was installed but still not detected.

Solution:
Activated virtual environment and installed packages again.

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3. Encoding Issue While Reading File

Problem:

Some special characters caused errors while reading the file.
Placed the file in the correct directory. Because default file encoding was not specified.

Solution:

Used UTF-8 encoding while opening file:

open("corpus.txt", "r", encoding="utf-8")

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4. Python Version Compatibility Issue

Error:
IndentationError inside torch library

Reason:
Python 3.13 is not fully compatible with PyTorch.

Solution:
Switched to Python 3.10 and created a new virtual environment.

Output

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Enter your search query (or type 'exit'): Artificial Intelligence

🔍 Top 3 Results:

Sentence: Reinforcement Learning from Human Feedback (RLHF) helps align AI with values.
Similarity Score: 0.4805
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Sentence: Artificial intelligence is transforming the way we write code in VS Code.
Similarity Score: 0.4751
--------------------------------------------------
Sentence: Neural networks are loosely inspired by the structure of the human brain.
Similarity Score: 0.3519
--------------------------------------------------
```

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Enter your search query (or type 'exit'): NLP

🔍 Top 3 Results:

Sentence: Sentiment analysis is a common NLP task used to track brand reputation online.
Similarity Score: 0.5030
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Sentence: BERT and RoBERTa are encoder-only models used for understanding text context.
Similarity Score: 0.3267
--------------------------------------------------
Sentence: Large Language Models require massive amounts of clean data for fine-tuning.
Similarity Score: 0.3253
--------------------------------------------------
```

Conclusion

This task helped me understand how semantic search works using embeddings instead of keyword matching. I learned how to convert text into vectors and use cosine similarity to find relevant results.